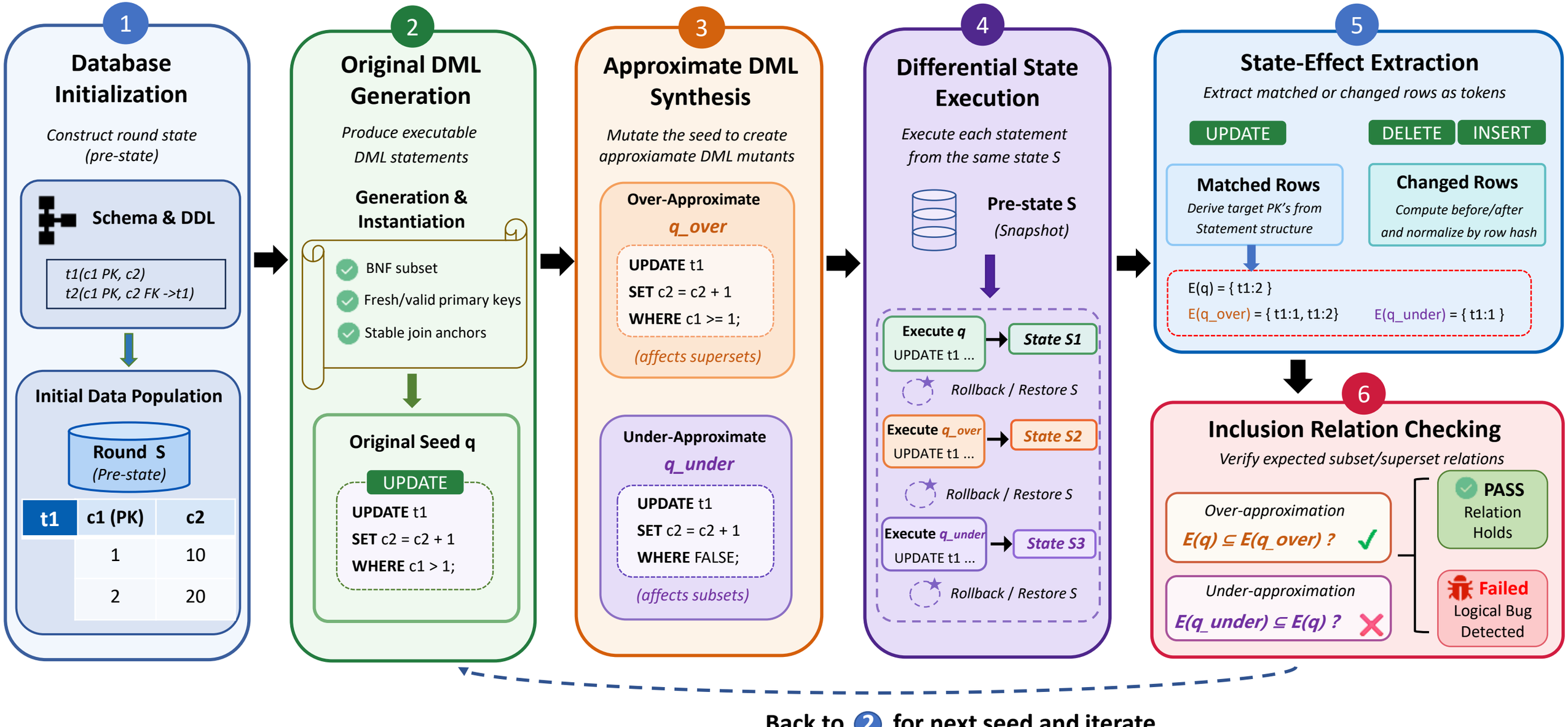
1
Database Initialization
Construct round state (pre-state)
Schema & DDL
t1(c1 PK, c2)
t2(c1 PK, c2 FK ->t1)
Initial Data Population
Round S (Pre-state)
t1 | c1 (PK) | c2
1 | 10
2 | 20
2
Original DML Generation
Produce executable DML statements
Generation & Instantiation
BNF subset
Fresh/valid primary keys
Stable join anchors
Original Seed q
UPDATE
UPDATE t1
SET c2 = c2 + 1
WHERE c1 > 1;
3
Approximate DML Synthesis
Mutate the seed to create approxiamate DML mutants
Over-Approximate
q_over
UPDATE t1
SET c2 = c2 + 1
WHERE c1 >= 1;
(affects supersets)
Under-Approximate
q_under
UPDATE t1
SET c2 = c2 + 1
WHERE FALSE;
(affects subsets)
4
Differential State Execution
Execute each statement from the same state S
Pre-state S
(Snapshot)
Execute q
UPDATE t1 ...
State S1
Rollback / Restore S
Execute q_over
UPDATE t1 ...
State S2
Rollback / Restore S
Execute q_under
UPDATE t1 ...
State S3
Rollback / Restore S
5
State-Effect Extraction
Extract matched or changed rows as tokens
UPDATE
DELETE
INSERT
Matched Rows
Derive target PK's from Statement structure
Changed Rows
Compute before/after and normalize by row hash
E(q) = { t1:2 }
E(q_over) = { t1:1, t1:2}
E(q_under) = { t1:1 }
6
Inclusion Relation Checking
Verify expected subset/superset relations
Over-approximation
E(q) ⊆ E(q_over) ?
Under-approximation
E(q_under) ⊆ E(q) ?
PASS
Relation Holds
Failed
Logical Bug Detected
Back to 2 for next seed and iterate